WellBot

AI-Powered Healthcare Chatbot

Your AI Wellness Partner, Always By Your Side

Presented by: Team-D

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Meet Team-D

M. Geethika

Backend Developer & AI Integration

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Database & Language switching in chatbot

Technical Stack

Frontend: HTML, CSS, React.js

Backend: Python, Flask API

AI/ML: Groq AI, RandomForest ML

Database: SQLite DB

Project Overview

WellBot is an AI-powered healthcare chatbot that provides personalized wellness guidance across multiple domains including mental health, fitness, nutrition, motivation, and stress relief.

Key Highlights:

- 6 Specialized AI Chat Specialists for different wellness areas
- AI Wellness Predictor powered by RandomForest ML model
- BMI Calculator for health metrics tracking
- Multi-language support (5+ languages) for global accessibility
- Real-time mood prediction and productivity scoring
- Interactive dashboard with weekly activity charts and wellness trends
- Chat export functionality for record keeping

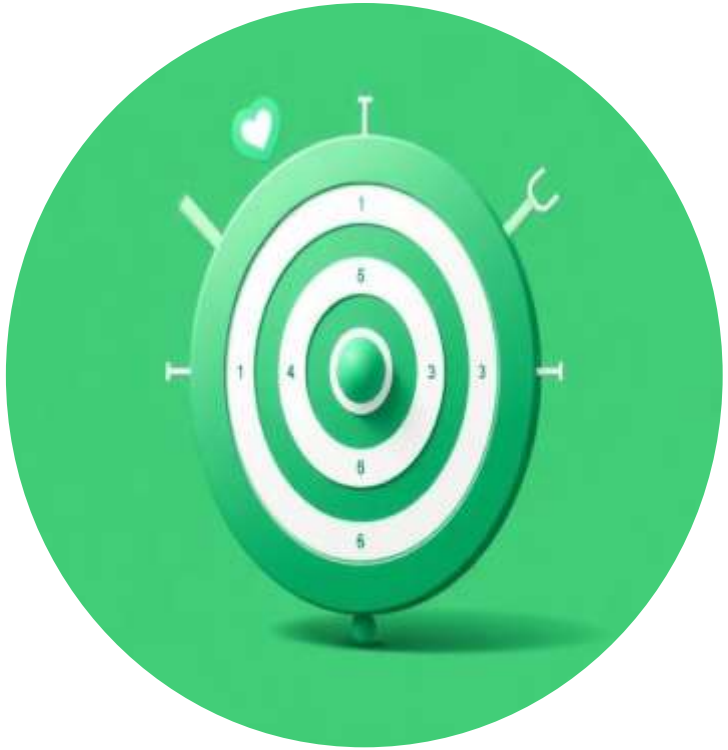
Vision & Mission

Our Vision

To revolutionize personal healthcare by making AI-driven wellness guidance accessible, affordable, and available 24/7 to everyone — breaking barriers of language, location, and cost.

Our Mission

- Provide expert-level wellness guidance through specialized AI chatbots
- Enable early wellness prediction using machine learning models
- Support multilingual conversations for inclusive healthcare
- Empower users to track and improve their daily wellness habits
- Deliver a seamless, user-friendly healthcare experience



Proposed Solution

AI Chat Specialists

6 domain-specific chatbots — Mental Health, Fitness Coach, Nutrition, Motivation, Stress Relief, and General Chat — each powered by Groq AI with tailored system prompts.

ML Wellness Predictor

RandomForest ML model that predicts mood, stress level, and productivity score based on daily inputs like sleep, exercise, caffeine, screen time, and more.

Multi-Language Support

14+ language support including Hindi, Telugu, Tamil, and more — AI responds in user's preferred language for inclusive healthcare access.

Smart Dashboard

Interactive dashboard with weekly activity charts, mood trends, streak tracking, and feedback ratings — giving users a holistic wellness overview.

Our Development Process

01

Research & Planning

Identified key wellness domains, studied existing healthcare chatbots, and defined project scope and requirements.

02

UI/UX Design

Designed dark-themed, modern UI with intuitive navigation, dashboard layout, and responsive chat interfaces.

03

Frontend Development

Built with HTML, CSS & React.js — creating reusable components, routing, and a seamless user experience.

04

Backend & AI Integration

Developed Flask API endpoints, integrated Groq AI for chat, trained RandomForest ML model for wellness prediction.

05

Testing & Deployment

Comprehensive testing across browsers, multilingual testing, performance optimization, and final deployment.

Application Workflow

1. User visits Landing Page
2. Sign Up / Login with credentials
3. Access Dashboard with wellness overview
4. Choose a Chat Specialist (Mental Health, Fitness, etc.)
5. Chat with AI in preferred language
6. Use AI Wellness Predictor (enter daily stats)
7. Get mood, stress & productivity predictions
8. Use BMI Calculator for health assessment
9. Track progress via charts & trends
10. Export chat history for records

System Architecture

Frontend Layer

HTML, CSS, React.js
Responsive UI, Dark Theme, Multi-language

API Layer (Flask)

RESTful API Endpoints
Authentication, Chat, Prediction, BMI

AI/ML Layer

Groq AI (Chat Engine)
RandomForest ML (Wellness Predictor)

Database Layer

SQLite DB
Users, Chats, Feedback, Wellness Data

Technical Stack

⚡ Frontend

- HTML5 & CSS3 — Structure and styling
- React.js — Component-based UI framework
- Responsive Design — Mobile & desktop compatible

⚡ Backend

- Python — Core programming language
- Flask — Lightweight web framework & REST API
- Flask-CORS — Cross-origin resource sharing

⚡ AI & ML

- Groq AI — Fast LLM inference for chat responses
- RandomForest ML — Wellness prediction model
- Scikit-learn — ML model training and evaluation

⚡ Database

- SQLite — Lightweight relational database
- User auth, chat history, feedback storage

BMI Calculator Feature

WellBot includes a built-in BMI (Body Mass Index) Calculator that helps users quickly assess their body weight category based on height and weight inputs.

How It Works:

1. User enters their height (cm) and weight (kg)
2. BMI is calculated using the standard formula: weight / (height^2)
3. Result is categorized: Underweight, Normal, Overweight, or Obese
4. Personalized health tips are provided based on the BMI category
5. Users can track BMI changes over time through the dashboard

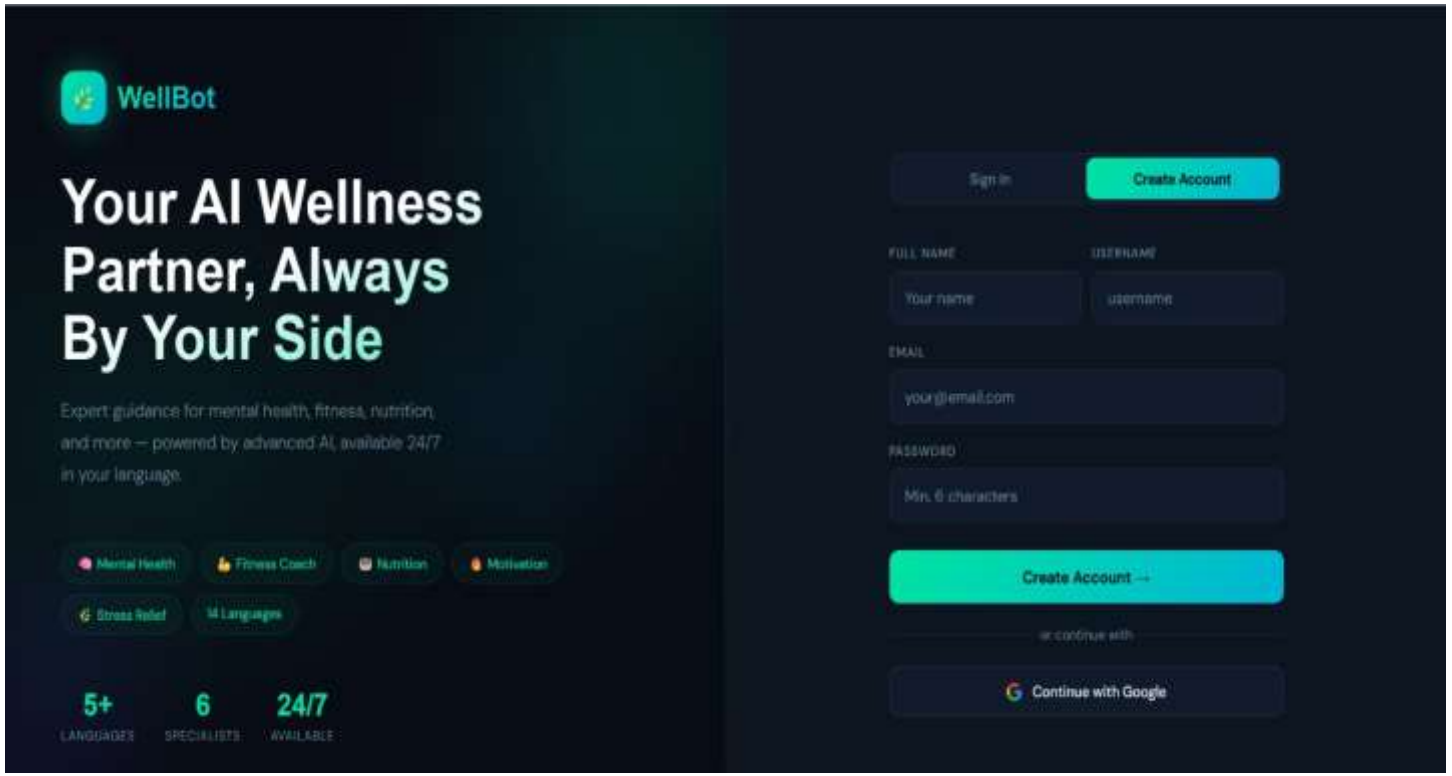
BMI is a quick health indicator that complements the AI Wellness Predictor for comprehensive health assessment.

Landing Page

Key Elements:

- Modern dark-themed design with green accents
- Hero section with clear value proposition
- Interactive dashboard preview showing stats
- Category badges for wellness domains
- Call-to-action buttons: 'Start for Free' & 'See Features'
- Navigation with Features, AI Engine, Specialists sections

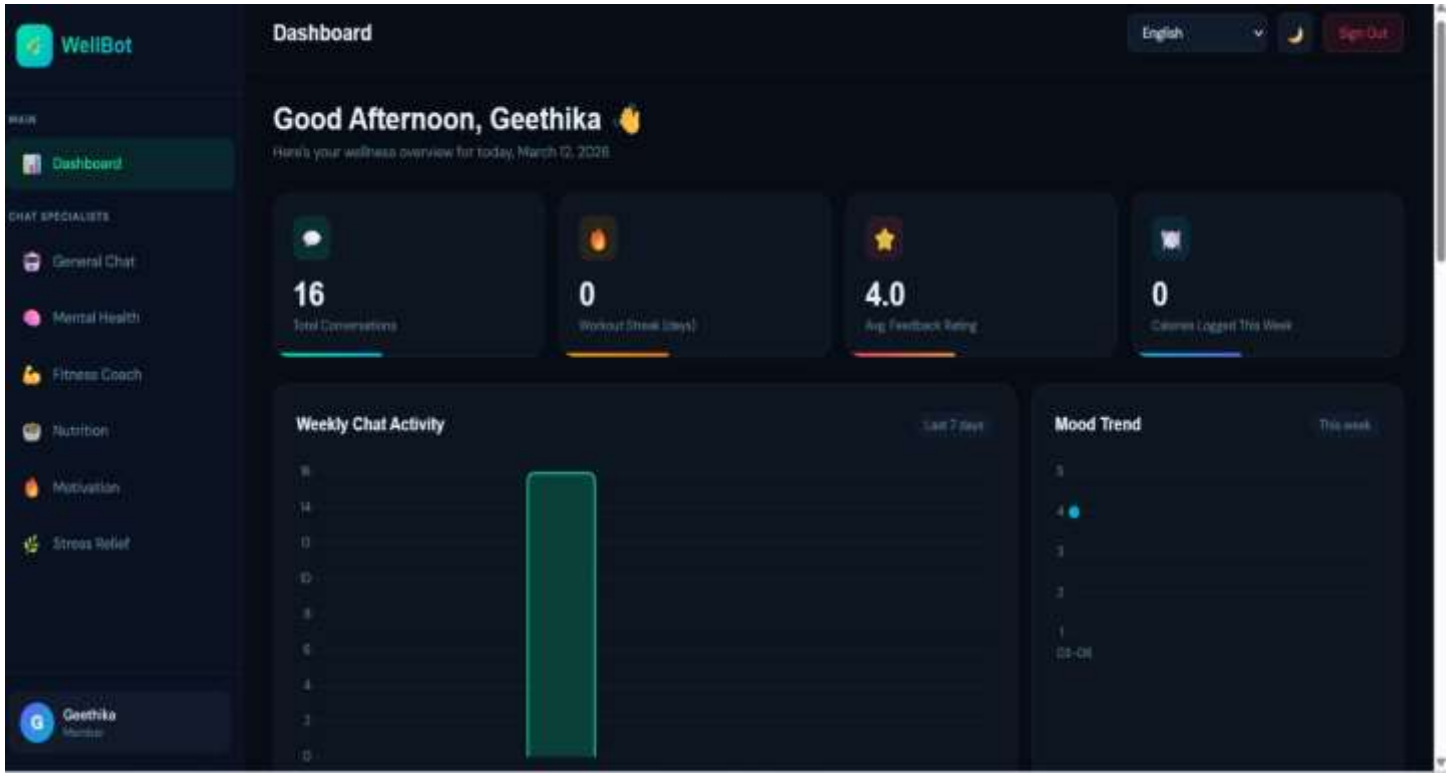
Login & Registration



Key Elements:

- Sign In / Create Account toggle tabs
- Full Name, Username, Email & Password fields
- 'Continue with Google' social login option
- Gradient CTA button for account creation
- Left panel shows platform features & stats
- Stats: 5+ Languages, 6 Specialists, 24/7 Available

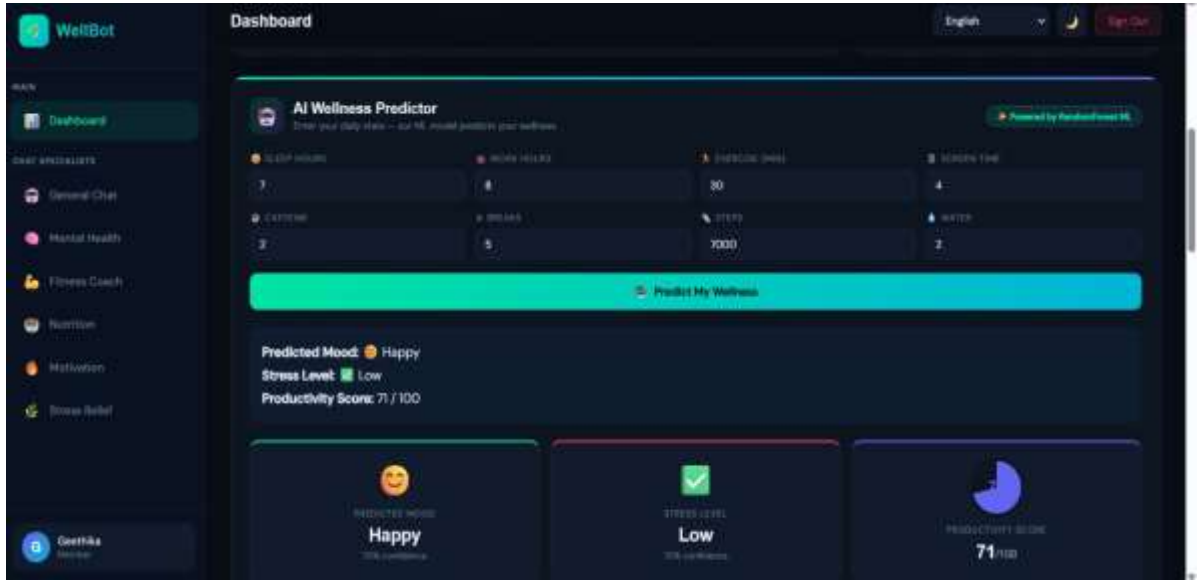
Dashboard & Analytics

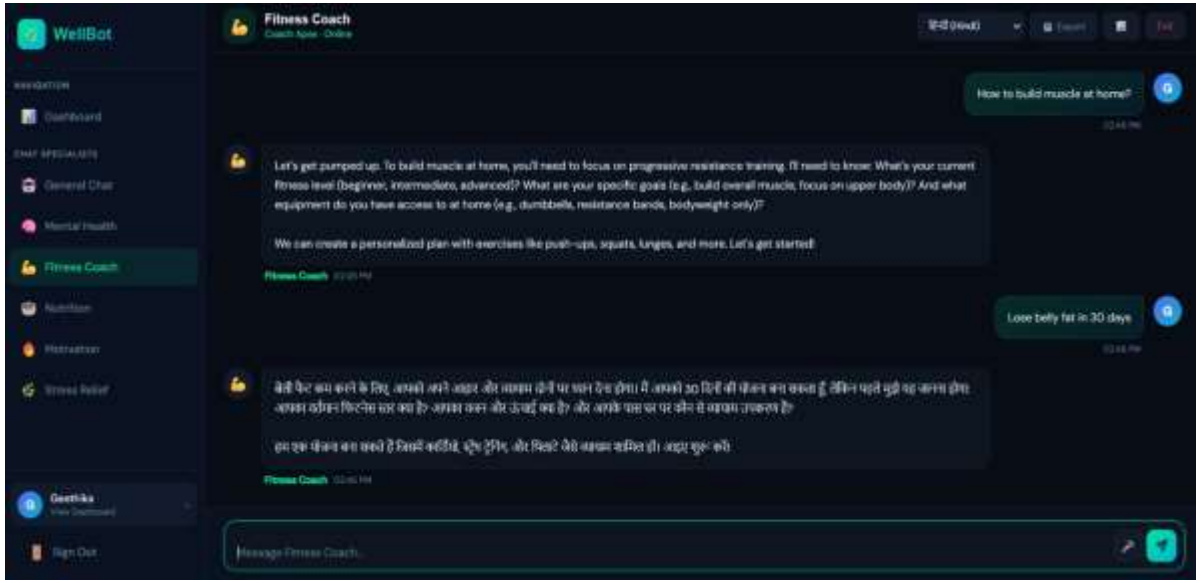


Key Elements:

- Personalized greeting with time-based message
- Stat cards: Conversations, Streak, Rating, Calories
- Weekly Chat Activity bar chart
- Mood Trend visualization over time
- Left sidebar with specialist navigation
- Language selector & dark mode toggle

AI Wellness Predictor & Chatbot





AI Wellness Predictor

- Enter daily stats: sleep, work, exercise, caffeine, steps, water, screen time, breaks
- ML model predicts Mood, Stress Level & Productivity Score
- Visual result cards with confidence percentages
- Powered by RandomForest ML algorithm

AI Chat Specialists

- 6 specialist chatbots for different wellness areas
- Real-time AI responses powered by Groq
- Multi-language support — responds in user's language
- Chat export and history tracking features

Feature Enhancements

Smart Notifications

Push notifications for hydration reminders, exercise breaks, and mood check-ins.

Mobile App (PWA)

Progressive Web App support for native mobile-like experience on smartphones.

Community Forum

Peer support groups and wellness communities for shared motivation and experiences.

Doctor Integration

Connect with real healthcare professionals for advanced medical consultations.

Advanced Analytics

Monthly wellness reports, trend analysis, and personalized improvement suggestions.

Gamification

Achievement badges, wellness challenges, leaderboards, and reward points system.

Wearable Integration

Sync with fitness trackers and smartwatches for automatic health data tracking.

Enhanced Security

Two-factor authentication, encrypted chats, and HIPAA-compliant data handling.

Thank You!

WellBot — Your AI Wellness Partner, Always By Your Side

WellBot demonstrates the power of combining AI, Machine Learning, and modern web technologies to create a comprehensive healthcare assistant. With 6 specialized chatbots, ML-powered wellness predictions, BMI calculator, and multi-language support — WellBot makes personalized healthcare guidance accessible to everyone, anytime, anywhere.

Team-D

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Questions? We'd love to hear from you! 🙌